

Homework Assignment: Logistic Regression

1 Learning objectives

- Understand odds and odds ratio
- Build logistic regression models
- Interpret coefficients

2 Problems

2.1 Problem 1: Conceptual Understanding of Logistic Regression

Suppose researchers are studying whether a biomarker X is associated with the presence of a disease. The outcome is coded as:

$$y = \begin{cases} 1 & \text{disease} \\ 0 & \text{no disease} \end{cases}$$

A logistic regression model is fitted:

$$\log\left(\frac{p}{1-p}\right) = -1.5 + 0.7X$$

where $p = P(Y = 1|X)$.

2.1.1 Part A

When $X = 0$, calculate:

1. the log-odds of disease
2. the odds of disease
3. the probability of disease

2.1.2 Part B

Calculate the odds ratio associated with a one-unit increase in X . Interpret this odds ratio in the context of the biological study.

2.1.3 Part C

Calculate the probability of disease for $X = 1$ and $X = 2$. Does increasing X by one unit produce the same absolute increase in probability each time? Explain why or why not.

2.1.4 Part D

Using a classification threshold of 0.5, determine the value of X at the decision boundary. For what values of X would the model predict disease?

2.1.5 Part E

A student interprets the coefficient 0.7 as:

“Every one-unit increase in the biomarker increases the probability of disease by 70%.”

Explain why this interpretation is incorrect and provide the correct interpretation.

2.1.6 Part F

Suppose X had first been standardized using `StandardScaler`. How would your interpretation of the coefficient 0.7 change?

2.2 Problem 2: Predicting Diabetes Using Logistic Regression

The Pima Indians Diabetes dataset in the `data` folder contains health measurements collected from adult women of Pima Indian heritage. The outcome indicates whether an individual was diagnosed with diabetes. In this problem, you will use logistic regression to investigate whether physiological measurements can be used to predict diabetes status and interpret how individual predictors are associated with the odds of diabetes.

$$\text{Outcome} = \begin{cases} 1 & \text{diabetes} \\ 0 & \text{no diabetes} \end{cases}$$

The predictor variables include measurements such as glucose concentration, blood pressure, BMI, age, and number of pregnancies.

2.2.1 Part A: Explore the biological data

Report the number of observations and predictor variables and the number of individuals with and without diabetes. Examine the distributions of the predictor variables. Identify any zero values that are biologically implausible. Replace these with NaN to consider them as missing values and impute them before applying `StandardScaler`. As such, your `Pipeline` needs to consist of three sequential steps: imputation, standardization, and logistic regression. It is worth noting that these three steps would be applied to the training data only.

2.2.2 Part B: Prepare the data for modeling

Divide the dataset into a 75% training set and 25% testing set, using a stratified split and a fixed `random_state`. Standardize the continuous predictors using `StandardScaler`. Make sure that any preprocessing steps that learn information from the data are fitted using the training data only.

Briefly explain why preprocessing the entire dataset before the train/test split could lead to data leakage.

2.2.3 Part C: Fit a logistic regression model

Using the training data, fit a logistic regression model to predict diabetes status.

Report the training and testing accuracy. Compare the two values and briefly discuss what they suggest about the model's ability to generalize to unseen observations.

2.2.4 Part D: Interpret predicted probabilities

Select three individuals from the test set:

- the individual with the lowest predicted probability of diabetes;
- the individual whose predicted probability is closest to 0.5; and
- the individual with the highest predicted probability of diabetes.

For each individual, report the predictor values, true diabetes status, predicted probability of diabetes, and predicted class using a threshold of 0.5.

Explain why the individual with a probability closest to 0.5 can be considered a relatively uncertain prediction.

2.2.5 Part E: Interpret the logistic regression coefficients

Extract the fitted coefficients θ_j . Because the predictors have been standardized, calculate

$$\text{odds ratio}_j = e^{\theta_j}$$

for each predictor.

Create a table with three columns: Predictor, Coefficient θ_j , Odds ratio e^{θ_j} . Identify the predictor with the largest positive coefficient and interpret its coefficient and odds ratio in the context of diabetes.

2.2.6 Part F: Scientific interpretation

Based on your analysis, write a short paragraph addressing these questions: Which physiological measurements appear most strongly associated with diabetes prediction in this model? Can you conclude that increasing or decreasing one of these measurements would cause a corresponding change in diabetes risk? Why or why not? What are two limitations you would consider before applying this model to a broader population?